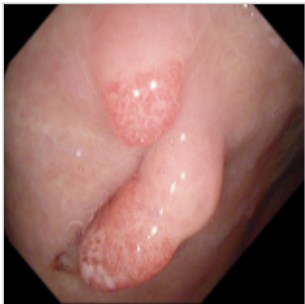
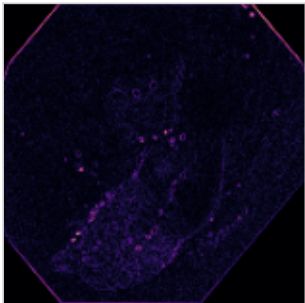


Haar wavelet decomposition and the WFM low-frequency style perturbation (real frame)

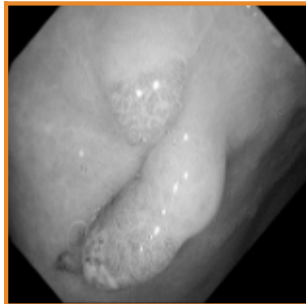
Input frame



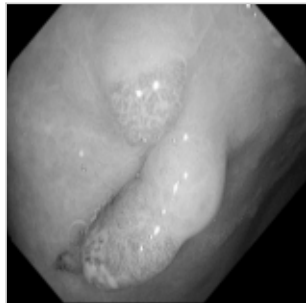
High-freq energy



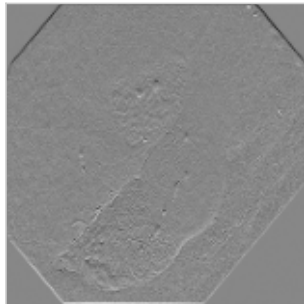
LL (low-freq: style)



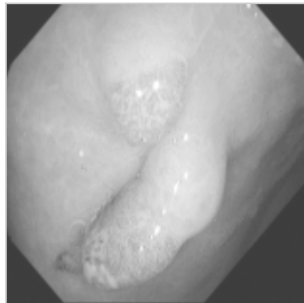
Recon. (baseline)



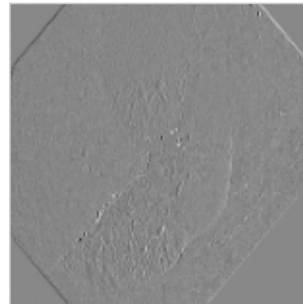
LH (horizontal)



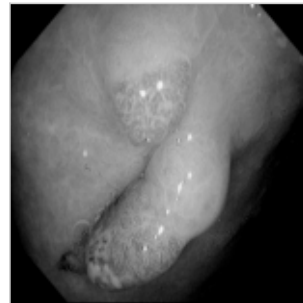
LL style A ($\mu\uparrow, \sigma\downarrow$)



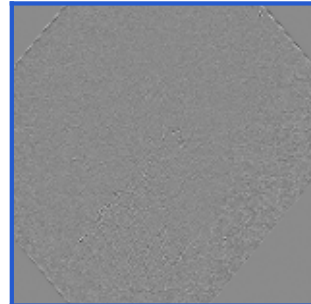
HL (vertical)



LL style B ($\mu\downarrow, \sigma\uparrow$)



HH (diagonal: edges)



Frequency-domain style perturbation

Low frequency (LL) carries the scanner / illumination style; high frequency (LH·HL·HH) carries polyp structure & edges.

Perturbing LL in training changes appearance, not shape $\rightarrow$ domain-invariant features.